Class-Guided Feature Decoupling Network for Airborne Image Segmentation

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***Abstract*— Contextual information has been demonstrated to be helpful for airborne image segmentation. However, most of the previous works focus on the exploitation of spatially contextual information, which is difficult to segment isolated objects, mainly surrounded by uncorrelated objects. To alleviate this issue, we attempt to take advantage of the co-occurrence relations between different classes of objects in the scene. Especially, similar to other works, convolutional features are first extracted to capture the spatially contextual information. Then, a feature decoupling module is designed to encode the class co-occurrence relations into the convolutional features; thus, the most discriminative features can be decoupled. Finally, the segmentation result is inferred from the decoupled features. The whole process is integrated to form an end-to-end network, named class-guided feature decoupling network (CGFDN). Experimental results on two widely used benchmark data sets show that CGFDN obtains competitive results (>90% overall accuracy (OA) on 5-cm-resolution Potsdam and >91% OA on 9-cm-resolution Vaihingen) in comparison with several state-of-the-art models.**

***Index Terms*— Airborne images, co-occurrence relations, contextual information, feature decoupling, semantic segmentation.**

I. Introduction

With the rapid development of sensor technology, the spatial resolution of airborne images is becoming much higher (e.g., 5–10 cm). Benefiting from such abundant spatial information, airborne images have been widely used in many applications, such as environmental monitoring [1], disaster relief [2], and urban change detection [3], [4]. For these applications, semantic segmentation, which aims at assigning a semantic label to each pixel in the image, is a vital and necessary prerequisite. Different from other images (e.g., the hyperspectral image with >1-m spatial resolution [5], [6]), the appearance of objects (e.g., car, building, and tree) in airborne images will be increasingly heterogeneous and easily confusing. This phenomenon could easily lead to high intraclass variance and low interclass variance, which makes semantic segmentation a challenging task. Early works [7], [8] for semantic segmentation of remote sensing images mainly use handcrafted features to construct classifiers. However, such features do not have enough discrimination capacity, leading to unsatisfactory segmentation results [9].

Convolutional neural network (CNN) [10] has made a significant breakthrough in computer vision tasks in recent years [11]–[16] because it can automatically learn high-level features from the image in an end-to-end way via its local connectivity and weight sharing properties [17]. As a kind of CNN, fully convolutional network (FCN)-based models have achieved much success in semantic segmentation. Long *et al.* [12] proposed an FCN architecture called FCN-8s, which combines the semantic information from a deep layer and the appearance information from a shallow layer to produce accurate and detailed segmentation results. In order to simultaneously capture the contextual and the location information, a symmetric architecture, consisting of a contracting path and an expansive path, was proposed in [18]. Badrinarayanan *et al.* [19] proposed an encoder–decoder architecture to retain the boundary information and save memory. More precisely, the decoder used the max-pooling indices in the encoder to upsample. Maggiori *et al.* [20] extracted features from different layers in FCN and concatenated them together to infer the final segmentation map. Marmanis *et al.* [21] and Volpi and Tuia [22] proposed to jointly learn a semantic segmentation task and a boundary detection task together with a multitask FCN architecture.

Although FCN-based models are able to capture the contextual information via convolutional operators, which is critical to the semantic segmentation of airborne images, the contextual information captured by convolutional operators is limited due to the limitation of the local receptive field. An intuitive method to address this issue is to increase the number of convolutional layers; thus, the deep CNN structures (e.g., VGGNet [23] and ResNet [24]) are often chosen as the backbone network. Nevertheless, it is shown that the empirical receptive field of CNNs is much smaller than the theoretical one, especially in deep layers [25]. Therefore, FCN-based models are difficult to capture sufficient contextual information as expected. Taking the top row in Fig. 1 as an example, although FCN-8s utilizes ResNet as the backbone network, it still fails to accurately distinguish the car in the purple box from the road.

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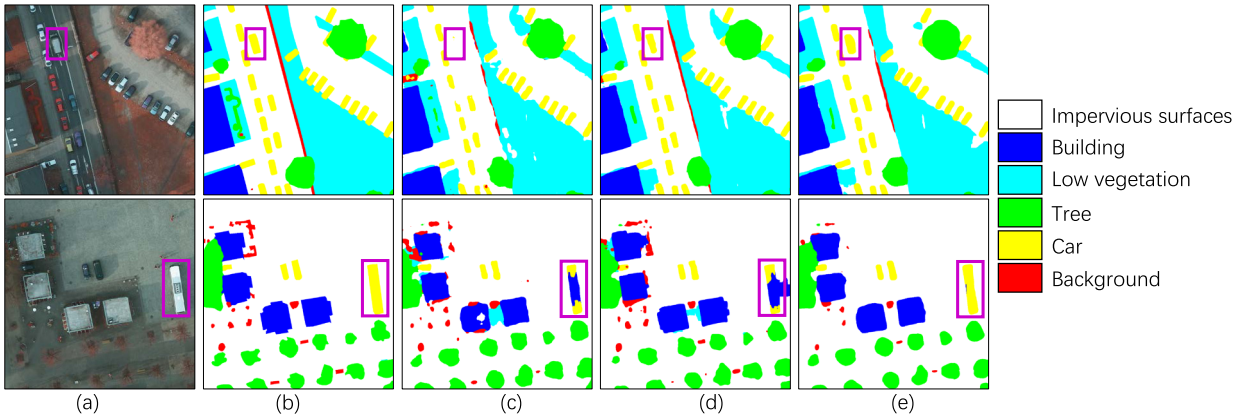


Fig. 1. Segmentation results obtained by three different models. (a) Airborne image. (b) Ground truth. (c) FCN-8s. (d) DANet. (e) CGFDN.

To capture richer contextual information, various strategies [26]–[32] have been explored, including multiscale context fusion, multilevel context fusion, and local-global context fusion. Audebert *et al.* [26] proposed to combine multiscale predictions via a multikernel convolutional layer composed of parallel convolutional kernels with different sizes. Liu *et al.* [27] proposed to aggregate multiscale contextual information obtained by different kernel sizes of dilated convolutions. Marcu and Leordeanu [28] designed a dual-stream network to learn local and global contextual information, respectively. Zhao *et al.* [29] proposed a pyramid pooling module to extract different subregions by adopting varying-size pooling kernels in a few strides. Chen *et al.* [30], [31] proposed an Atrous spatial pyramid pooling (ASPP) that explored convolutional features at multiple scales to encode global context information. Fu *et al.* [32] proposed a dual-attention network (DANet) to adaptively integrate local features with their global dependencies via a self-attention mechanism.

Compared with the original FCN-based models, the aforementioned models can make use of richer contextual information, thus improving the segmentation performance. For example, in the top row of Fig. 1, DANet is able to precisely segment the car, which is misclassified by FCN-8s. However, similar to the FCN-based models, all these models mainly focus on the exploitation of spatially contextual information, which is difficult to segment isolated objects, mainly surrounded by uncorrelated objects. As shown in the bottom row of Fig. 1, the pixels labeled as "car" in the purple box are surrounded by pixels from the class of "building." Both FCN-8s and DANet misclassify a large portion of the "car" as the "building," which is in contradiction with the common knowledge that the building is never over the car because they only took account of spatially contextual information.

In order to alleviate the above issue, we propose to resort to the co-occurrence relations between different classes of objects in the scene. For instance, "car" and "road" appear together with a high probability, but it is impossible to make "car" and "roof" appear together with a high probability. Such co-occurrence information can improve segmentation performance. Along this direction, we design a feature decoupling module to encode the co-occurrence information and further enhance the extracted features from spatial relationships solely. Especially, ResNet18 [24] is adopted as the backbone network to extract convolutional features. Then, these features are fed into our proposed decoupling module to obtain enhanced representations. Finally, the segmentation result is derived from these representations. The whole process is integrated to form an end-to-end network, named class-guided feature decoupling network (CGFDN). Fig. 1 demonstrates a segmentation example of CGFDN in the bottom line. As expected, the "car" in the purple box is precisely located and classified due to the incorporation of co-occurrence information.

In summary, the main contributions of our work are as follows.

1) We propose a novel CGFDN for airborne image segmentation. In addition to the use of spatially contextual information, we also take advantage of the co-occurrence relations between different classes of objects, thus improving the segmentation performance, especially for the isolated objects in the scene.
2) We design a feature decoupling module to encode the co-occurrence relations and further enhance the discriminative ability of the convolutional features. In order to achieve this goal, we consider making use of the predefined label information, whose representation can inherently contain the co-occurrence property.
3) Comprehensive experiments are conducted on two benchmark data sets. The results show that the co-occurrence relations can be captured as expected and the performance can be improved, which sufficiently validates the effectiveness of our proposed model.

II. CGFDN

The overall framework of CGFDN is shown in Fig. 2. It mainly consists of three steps. First, the airborne image is fed into a backbone network ResNet18 [24] to extract convolutional features. Similar to [24], our ResNet18 is also comprised of five convolutional layers (i.e., Conv1, Conv2, Conv3, Conv4, and Conv5), but the max-pooling operator in the second layer is excluded to retain more spatial information. Second, the last three convolutional layers are separately

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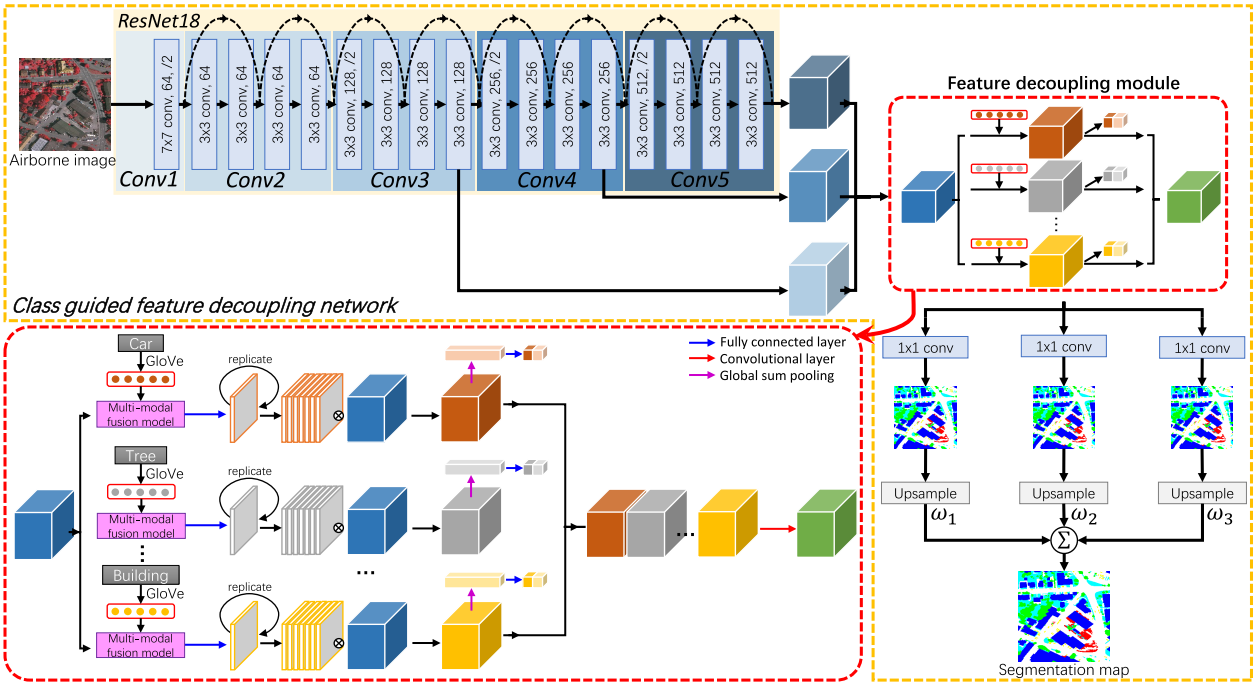


Fig. 2. Framework of the proposed CGFDN.

inputted into the feature decoupling module. This module aims at encoding the co-occurrence relations between different classes of objects to enhance the discriminative ability of the input convolutional feature. Finally, the segmentation map is inferred from the enhanced feature via a 1×1 convolutional layer and an upsampling operator. To capture richer contextual information, a weighted summation method is adopted to fuse the three segmentation maps. All the parameters in CGFDN can be adaptively learned from data in an end-to-end manner.

A. Feature Decoupling Module

As shown in Fig. 2, the key component of CGFDN is the feature decoupling module. Assume that the extracted convolutional feature is denoted as $\mathbf{F} \in \mathbb{R}^{C \times H \times W}$, where H, W, and C represent height, width, and channel numbers, respectively. For each pixel in $\mathbf{F}$, it can be considered as a feature representation of a local patch in the input airborne image. Therefore, all the spatial information in the patch will help the classification of this pixel, leading to satisfactory performance in most cases. However, for some isolated objects, their neighboring regions contain less information that is useful for classification. To alleviate this issue, our decoupling module attempts to take advantage of the co-occurrence relations between different classes of objects. As we all know, some objects often appear together with other ones. We argue that such co-occurrence information is helpful for the segmentation task. Nevertheless, how to effectively encode the co-occurrence information into $\mathbf{F}$ is also a challenging problem.

In addition to the convolutional features, another information that we can use is the predefined class labels, such as "car" and "tree." Benefiting from the development of natural language processing, each label can be encoded as a feature vector. More importantly, some techniques, such as GloVe [33], are capable of considering the co-occurrence relations between different words during their coding processes, which can be naturally matched with our purpose. Based on this motivation, our decoupling module takes the label information along with the convolutional features as inputs and aims to learn an enhanced feature representation.

For the kth class label, $k \in \{1, 2, \ldots, n\}$, where n is the number of predefined class labels, we apply a pretrained GloVe model using the Wikipedia data set to encode it into a D-dimensional feature vector $o_k \in \mathbb{R}^D$. Then, $\mathbf{F}$ and o_k are fed into a multimodal fusion model to derive an attention map $\mathbf{A}_k$. This attention map is expected to highlight the features in $\mathbf{F}$, which has large contributions to the kth class of objects or its co-occurrence ones. Especially, o_k is first expanded to a 3-D tensor $\mathbf{O}_k \in \mathbb{R}^{D \times H \times W}$, where the values in the ith channel are equal to the ith element of o_k. Then, inspired from [34], $\mathbf{O}_k$ and $\mathbf{F}$ can be projected into a space, which can be formulated as follows:

$$\mathbf{R}_k = \mathbf{P}^{\mathrm{T}} \sigma(\mathbf{U}^{\mathrm{T}}\mathbf{F} \circ \mathbf{V}^{\mathrm{T}}\mathbf{O}_k) + b \tag{1}$$

where $\mathbf{R}_k \in \mathbb{R}^{Q \times H \times W}$ is the projection result, $\mathbf{P} \in \mathbb{R}^{D' \times Q}$, $\mathbf{U} \in \mathbb{R}^{C \times D'}$, and $\mathbf{V} \in \mathbb{R}^{D \times D'}$ are projection matrices, $b \in \mathbb{R}^Q$ is a bias vector, and "σ" and "$\circ$" denote the nonlinear activation function and the Hadamard product operator, respectively. D' and Q represent the numbers of channels of the intermediate projection and the final projection, respectively. Finally, the attention map $\mathbf{A}_k$ can be derived as

$$\mathbf{A}_k = \mathrm{Softmax}(w\mathbf{R}_k + b_1) \tag{2}$$

where $w \in \mathbb{R}^Q$ and $b_1 \in \mathbb{R}$ represent the weight and the bias, respectively. Note that all the computations in (1) and (2) can

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TABLE I
DETAILED CONFIGURATION OF THREE FEATURE DECOUPLING MODULES

Operator \ Input layer	*Conv3*	*Conv4*	*Conv5*
$FC(\mathbf{P})$	64×64	128×128	256×256
$FC(\mathbf{U})$	128×64	256×128	512×256
$FC(\mathbf{V})$	300×64	300×128	300×256
$FC(b)$	64	128	256
$FC(w)$	64	128	256
$FC(b_1)$	1	1	1
$FC(\mathbf{W})$	2×128	2×256	2×512
$FC(b_2)$	2	2	2

be implemented by the fully connected operators so that they could be easily plugged into the whole network for end-to-end learning.

Once the attention map $\mathbf{A}_k$ is acquired, we replicate it C times along the channel dimension so that it can have the same size as $\mathbf{F}$. Since $\mathbf{A}_k$ contains larger values at the positions that have co-occurrence relations with the kth class of objects, multiplying it with $\mathbf{F}$ will decouple the most discriminative features $\mathbf{F}_k \in \mathbb{R}^{C\times H\times W}$. Considering that the total number of predefined class labels is n, we will generate n different discrimination features. To fuse them together, we first concatenate them into a high-dimensional tensor along the channel direction and then use a 1×1 convolutional layer to derive the final feature representation $\mathbf{G} \in \mathbb{R}^{C\times H\times W}$. Due to the incorporation of co-occurrence information, $\mathbf{G}$ enhances the discriminative ability of $\mathbf{F}$. It is worth noting that we make use of three different convolutional layers in Fig. 2. Therefore, three decoupling modules are designed, which has the same structure but with different numbers of parameters. The detailed explanations about their configurations are demonstrated in Table I, where "FC" represents the fully connected operator. In this article, D is empirically set to 300, and C is empirically set to 128, 256, and 512 for Conv3, Conv4, and Conv5, respectively. In addition, both D' and Q are set to 64, 128, and 256 for three feature decoupling modules related to Conv3, Conv4, and Conv5, respectively.

B. Loss Function

To ensure that the feature $\mathbf{F}_k \in \mathbb{R}^{C\times H\times W}$ has enough discriminative ability, we design a dc loss to supervise the learning process of each feature decoupling module directly. Especially, we apply a global summation pooling (GSP) operator to $\mathbf{F}_k$ and then feed the result into an output layer to get a 2-D output vector $\hat{y}_k$. The whole process can be formulated as

$$\hat{y}_k = \text{Sigmoid}(\mathbf{W}(\text{GSP}(\mathbf{F}_k)) + b_2) \tag{3}$$

where $\mathbf{W} \in \mathbb{R}^{2\times C}$ and $b_2 \in \mathbb{R}^2$ are weight and bias, respectively. Assume that the ground truth for the kth class is y_k, and the dc loss over all the classes for the lth decoupling module can be computed by

$$L_{\text{DC}}^l = \sum_{k=1}^{n} \left(y_k \log \hat{y}_k^l + (1 - y_k) \log\left(1 - \hat{y}_k^l\right)\right). \tag{4}$$

Besides the dc loss, we also use a segmentation loss to supervise the learning process of the whole framework. In particular, we apply the cross-entropy losses over the three segmentation maps and their fused one separately, which can be defined as

$$L_{\text{SG}} = \sum_{l=1}^{3}\sum_{k=1}^{n} \mathbf{S}_k \log \mathbf{Y}_k^l + \sum_{k=1}^{n} \mathbf{S}_k \log \mathbf{Y}_k \tag{5}$$

where $\mathbf{S}_k$, $\mathbf{Y}_k^l$, and $\mathbf{Y}_k$ denote the ground-truth map, the upsampled segmentation map for the lth module, and the fusion segmentation map, respectively. Combining the introduced dc loss and the segmentation loss together, the final loss of CGFDN is defined as

$$L = L_{\text{SG}} + \lambda\left(\frac{1}{3}\sum_{l=1}^{3} L_{\text{DC}}^l\right) \tag{6}$$

where λ is used to balance L_{SG} and L_{dc}.

III. EXPERIMENTAL RESULTS

A. Data Set Description

We evaluate the proposed CGFDN on two challenging benchmarks: Potsdam and Vaihingen, provided by Commission III of the ISPRS.[1]

1) Potsdam Data Set[2]*:* This data set contains 38 tiles with a spatial resolution of 5 cm. The size of all tiles is 6000×6000 pixels. Each tile consists of four channels (near-infrared, red, green, and blue). The corresponding digital surface model (DSM) and normalized DSM (nDSM) of each tile are also available. All tiles are densely annotated with six classes: impervious surface, building, low vegetation, tree, car, and background. Following the settings by Commission III of the ISPRS, we also utilize 24 images for training and the remaining 14 images for testing.

2) Vaihingen Data Set[3]*:* It contains 33 tiles with an average size of 2494×2064 pixels. The spatial resolution of this data set is 9 cm. Each tile is made up of three channels (near-infrared, red, and green). Different from the Potsdam data set, only DSM is available on the official website. Similar to the Potsdam data set, there are six classes (including impervious surface, building, low vegetation, tree, car, and background) in the Vaihingen data set. Following the settings by Commission III of the ISPRS, we use 16 images for training and the remaining 17 images for testing.

B. Implementation Details

For each image, we compute the normalized difference vegetation index (NDVI) as

$$\text{NDVI} = \frac{\text{IR} - R}{\text{IR} + R} \tag{7}$$

where IR and R denote the near-infrared channel and the red channel, respectively. Then, we build a composite input image consisting of the near-infrared, red, green, DSM, nDSM, and

[1] http://www2.isprs.org/commissions/comm3/wg4/semantic-labeling.html
[2] http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html
[3] http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html

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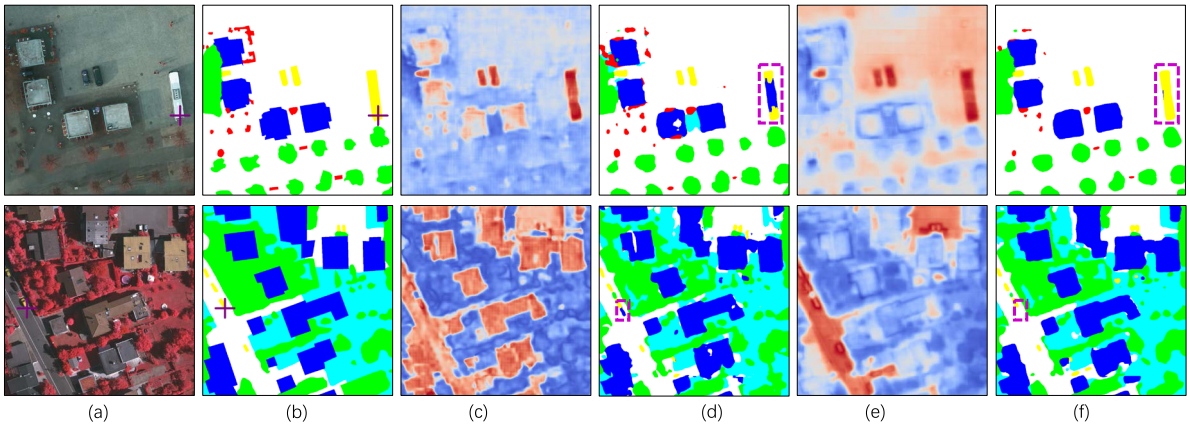


Fig. 3. Visualization of a given pixel achieved by the baseline model FCN-8s and CGFDN. Two pixels from the car (the first row) and impervious surface (the second row) are marked with purple cross. The similarity between all pixels and the selected one is normalized to the range [0, 1]. Blue: similarity values smaller than 0.5. Red: similarity values larger than 0.5. (a) Image. (b) Ground truth. (c) Baseline (context). (d) Baseline. (e) CGFDN (context). (f) CGFDN.

NDVI channels. For the Vaihingen data set, we refer to the released nDSM provided in [35]. To make full use of the GPU memory and generate as many training samples as possible, we use a 256×256 sliding window with a stride of 64 to crop the training samples from the training images. During testing, we crop the patches from each testing image using the same sliding method and process them one by one from the top left to the bottom right. For the overlapping regions, we use the averaged prediction value as the final result. To assess the performance of the proposed method, three widely used metrics are introduced. They are the overall accuracy (OA), F1 score, and mean of classwise intersection over union (mean IoU). OA is the metric that considers the global aspects of classification. F1 score is calculated as the harmonic mean of precision and recall. IoU is the overlapping rate of the predicted segmentation map and the ground-truth map.

We implement CGFDN with PyTorch [36]. The experiment is conducted on a 64-bit i7-4790 computer with 32-GB memory. One GeForce GTX TITAN X with 12-GB memory, under a 10.0 CUDA version, is employed in the experiment. Besides, we use Adam [37] to train CGFDN, and the parameters of Adam are set as default: $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e^{-8}$. The learning rate is initialized as $1e^{-3}$.

C. Ablation Study

The proposed CGFDN is constructed by plugging the feature decoupling module into FCN-8s; thus, we first compare with FCN-8s to evaluate the contributions of the proposed module. For a fair comparison, FCN-8s also takes ResNet18 as the backbone network. As shown in Table II, integrating the feature decoupling module improves the performance from 87.8% to 89.6% (1.8% ↑) on Potsdam and from 88.3% to 90.2% (1.9% ↑) on Vaihingen. Besides, different from FCN-8s, we use the weighted summation to fuse the segmentation results from the last three layers. To validate the effectiveness of the weighted summation method, we compare two CGFDNs with and without the weighted summation. The experimental results listed in Table II ("WS" denotes the weighted summation) demonstrate that the weighted summation method

TABLE II
PERFORMANCE (OA) COMPARISON OF DIFFERENT SETUPS OF CGFDN ON TWO BENCHMARK DATA SETS

Method	Input	WS	Aggregation	Potsdam	Vaihingen
FCN-8s	IRRG	*w/o*	-	87.8	88.3
CGFDN	IRRG	*w/o*	Concat	89.6	90.2
CGFDN	IRRG	*w*	Concat	90.3	90.6
CGFDN	IRRG	*w*	Sum	90.2	90.0
CGFDN	Composite	*w*	Concat	90.8	91.0

improves OA from 89.6% to 90.3% (0.7% ↑) on Potsdam and from 90.2% to 90.6% (0.4% ↑) on Vaihingen. As described in Section II-A, $\mathbf{G}$ is generated by a concatenation method. We denote this fusion procedure as "Concat" for simplicity. Apart from "Concat," we also try another way called "Sum" to directly fuse multiple class-specific features, which can be formulated as $\mathbf{G} = \sum_{k=1}^{n} \mathbf{F}_k$. The comparison results are shown in Table II. Compared with "Concat," "Sum" decreases the OA to 90.2% (0.1% ↓) on Potsdam and to 90.0% (0.6% ↓) on Vaihingen. Therefore, the "Concat" strategy is more suitable to our task. Furthermore, we investigate the effectiveness of DSM, nDSM, and NDVI by taking a composite image as the input. DSM and nDSM contain the height information, which is of the first importance to distinguish the confusing objects, such as roads/buildings and vegetation/trees. In addition, NDVI is a good indicator for vegetationm and it can further help to classify vegetation. By concatenating these additional data and IRRG together, CGFDN can improve the performance from 90.3% to 90.8% (0.5% ↑) on Potsdam and from 90.6% to 91.0% (0.4% ↑) on Vaihingen.

1) Context Visualization: In order to better understand how the feature decoupling module improves the segmentation result, we visualize the cosine similarity map between a given pixel and all the other pixels in the feature map. Similar to [41], this similarity map serves as a surrogate of the true object context. As shown in Fig. 3, we select two pixels from the classes of "car" and "impervious surface," respectively. For FCN-8s and CGFDN, we use the feature before the last softmax layer to calculate the feature similarity. For the first example, the feature of "car" is similar to that of "building"

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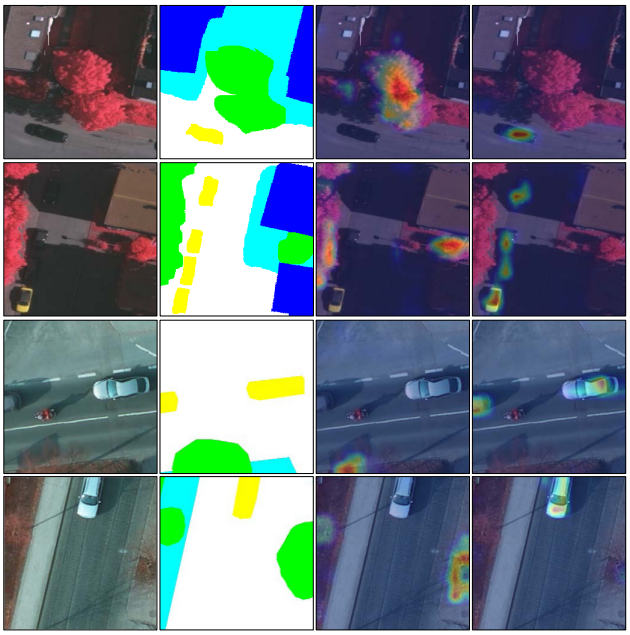


Fig. 4. Visualization of the airborne image, the corresponding ground truth, and two channel maps of the class-specific features. (a) Airborne image. (b) Ground truth. (c) Feature (tree). (d) Feature (car).

in the context map obtained by FCN-8s due to similar appearance, leading to low interclass variance. As a result, FCN-8s misclassifies the "car" into the "building" in the area marked by the purple dotted box. Different from FCN-8s, CGFDN obtains the enhanced representation after integrating the feature decoupling module into the baseline network. As can be seen in the context map of CGFDN, the feature of "car" is relevant to that of the class "impervious surface" due to their co-occurrence relations in the scene (i.e., the car is parked on the road). More importantly, this contextual information makes the features between "car" and "building" more distinguishable (features between the car and the building have no relevance to each other shown in the context map of CGFDN), leading to correct labeling of the car. For the second example, even if the road appears similar to the roof of the building, CGFDN is able to segment "impervious surface" correctly due to the use of the co-occurrence information, while FCN-8s fails.

2) Class-Specific Features Visualization: For class-specific features, it is hard to give comprehensible visualization directly. Instead, we show some attentional channels to see whether they highlight the desired semantic areas. As can be seen in Fig. 4, we find that the response of the specific semantic (i.e., tree and car) is noticeable after the use of the feature decoupling module. It is remarkable that the module could locate the cars occluded by the building shadow, as shown in the second row of Fig. 4. In summary, these visualization results further demonstrate that the proposed feature decoupling module can effectively locate the class-specific regions on the feature map.

3) Effects of Different λ Values: The introduced dc loss aims at supervising the learning process of the feature decoupling module. To demonstrate its effectiveness,

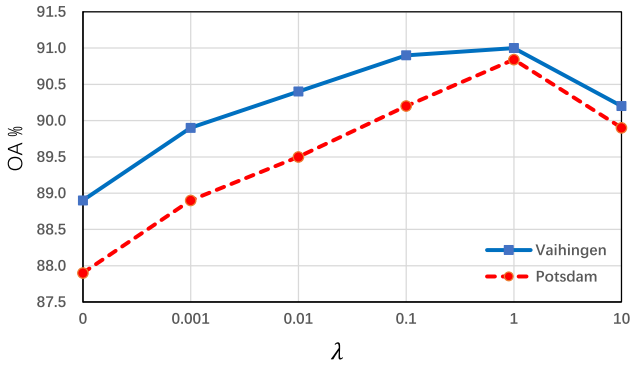


Fig. 5. Performance of CGFDN with different λ values.

we change the weight parameter λ from a candidate set of $\{0, 0.001, 0.01, 0.1, 1, 10\}$. As shown in Fig. 5, CGFDN obtains the lowest OA when $\lambda = 0$ (i.e., without dc loss) on Potsdam and Vaihingen. We can also observe that OA will first increase and then decrease with the increase of λ. Besides, CGFDN achieves the best OA on Potsdam and Vaihingen when $\lambda = 1$. Thus, λ is set to 1 in the following experiments.

D. Comparison With Models Using Spatially Contextual Information

To demonstrate the effectiveness of CGFDN, we compare it with several state-of-the-art deep learning models based on spatially contextual information on two benchmarks. The comparison models include FCN-8s [12], UNet [18], SegNet [19], PSPNet [29], DeepLabv3+ [31], PSANet [38], EncNet [39], S-RA-FCN [40], and DANet [32]. For these models, we take the same configuration provided by the authors. Note that we choose ResNet18 as the backbone network in FCN-8s, PSPNet, PSANet, EncNet, S-RA-FCN, and DANet for a fair comparison. All the deep learning methods take IRRG as the input.

Table III reports the quantitative performance of different models in terms of the OA, F1 score, and mean IoU on the Potsdam data set. As shown in this table, FCN-based models (e.g., FCN-8s, UNet, and SegNet) achieve the worst performance due to the lack of sufficient contextual information, while PSPNet improves the performance via a multiscale context fusion strategy. Similarly, PSANet and S-RA-FCN are able to aggregate long-range spatial relations among pixels, thus achieving better performance than FCN-based models. Compared with these three models (i.e., PSPNet, PSANet, and S-RA-FCN), EncNet performs better in terms of OA and mean F1 score because it fully utilizes the strong correlation between scene context and the probabilities of categories. Nevertheless, EncNet still obtains slightly worse performance than DeepLabv3+ due to the use of features from the last layer of the backbone network, which misses the detailed information related to object boundaries. To address this issue, DeepLabv3+ designs a decoder module to recover the object boundaries. However, the performance of EncNet and DeepLabv3+ is inferior to that of DANet. This phenomenon is due to a lack of the ability of leveraging the

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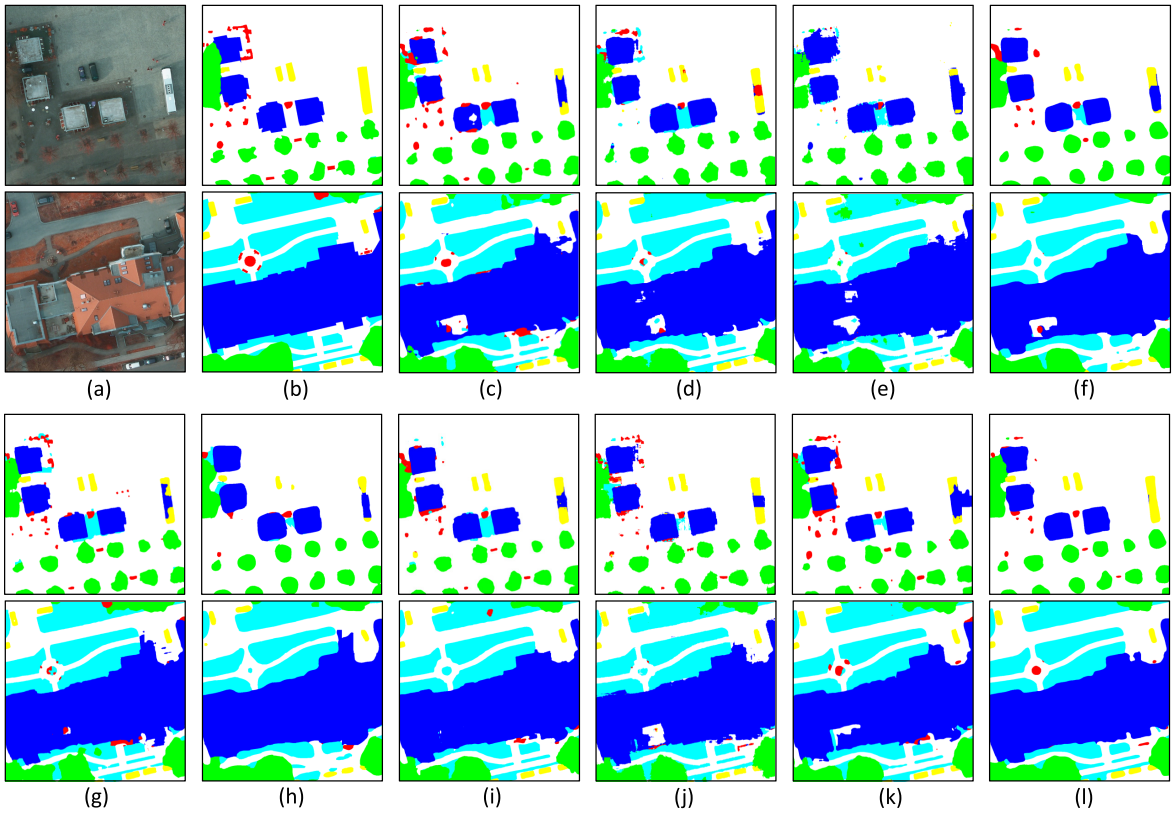


Fig. 6. Qualitative comparisons between different models using spatially contextual information on the Potsdam data set. Six classes are included in the segmentation maps: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red). (a) Image. (b) Ground truth. (c) FCN-8s. (d) UNet. (e) SegNet. (f) PSPNet. (g) DeepLabv3+. (h) PSANet. (i) EncNet. (j) S-RA-FCN. (k) DANet. (l) CGFDN.

TABLE III
PERFORMANCE ACHIEVED BY DIFFERENT MODELS USING SPATIALLY CONTEXTUAL INFORMATION ON THE POTSDAM DATA SET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1	Mean IoU
FCN-8s[1](ResNet18) [12]	89.9	93.7	83.0	85.2	93.5	87.8	89.1	71.7
UNet [18]	88.2	91.1	82.8	84.9	91.6	86.2	87.7	68.5
SegNet [19]	87.8	90.7	81.0	84.7	89.7	85.7	86.8	66.2
PSPNet[1](ResNet18) [29]	90.7	94.8	84.1	85.9	90.5	88.7	89.2	71.7
DeepLabv3+ [31]	91.3	94.8	84.2	86.6	93.8	89.2	90.1	73.8
PSANet[1](ResNet18) [38]	90.4	94.5	83.6	85.8	84.6	88.5	87.8	70.3
EncNet[1](ResNet18) [39]	91.0	94.9	84.4	85.9	93.6	89.0	90.0	73.2
S-RA-FCN[1](ResNet18) [40]	90.7	94.2	83.8	85.8	93.6	88.5	89.6	72.5
DANet[1](ResNet18) [32]	91.6	95.4	85.1	86.8	94.2	89.7	90.6	74.8
CGFDN	92.1	95.6	86.3	87.9	94.9	90.3	91.4	75.6

[1] We choose ResNet18 as the backbone network for fair comparison.

relationship between different objects in a global view in EncNet and DeepLabv3+. To remedy this, DANet introduces a self-attention mechanism to adaptively integrate local features with their global dependencies, thus obtaining better OA, mean F1 score, and mean IoU than EncNet and DeepLabv3+. In addition, the proposed CGFDN outperforms all comparison models using spatially contextual information. More specifically, CGFDN further improves OA from 89.7% to 90.3%, mean F1 score from 90.6% to 91.4%, and mean IoU from 74.8% to 75.6% compared with the second-best model DANet. As analyzed in Section I, spatially contextual information is insufficient to deal with the isolated objects, easily leading to misclassification. Different from the abovementioned comparison models, CGFDN attempts to utilize co-occurrence relations between different classes of objects to enhance the discriminative ability of the spatially contextual information. Fig. 6 shows the segmentation maps achieved by seven different models on the Potsdam data set. It can be observed that CGFDN obtains a more coherent and accurate segmentation map, especially for the isolated objects, than other models. Similar results are demonstrated in Table IV and Fig. 7 on the Vaihingen data set. Again, CGFDN, DANet, and DeepLabv3+ achieve better performance than other models. Especially, OA and mean F1 score obtained by CGFDN and DANet are higher than DeepLabv3+, and CGFDN obtains better performance than DANet and DeepLabv3+.

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TABLE IV

PERFORMANCE ACHIEVED BY DIFFERENT MODELS USING SPATIALLY CONTEXTUAL INFORMATION ON THE VAIHINGEN DATA SET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1	Mean IoU
FCN-8s[1](ResNet18) [12]	90.0	93.0	77.7	86.5	80.4	88.3	85.5	75.5
UNet [18]	90.5	93.3	79.6	87.5	76.4	89.2	85.5	75.5
SegNet [19]	90.2	93.7	78.5	85.8	83.9	88.5	86.4	76.8
PSPNet[1](ResNet18) [29]	90.6	94.3	79.0	87.0	70.7	89.1	84.3	74.1
DeepLabv3+ [31]	91.4	94.7	79.6	87.6	85.8	89.9	87.8	79.0
PSANet[1](ResNet18) [38]	90.1	93.7	79.2	87.5	67.3	88.9	83.6	73.1
EncNet[1](ResNet18) [39]	91.2	94.1	79.2	86.9	83.7	89.4	87.0	77.8
S-RA-FCN[1](ResNet18) [40]	90.5	93.8	79.6	87.5	82.6	89.2	86.8	77.3
DANet[1](ResNet18) [32]	91.0	94.5	80.9	88.1	83.9	90.0	87.7	78.7
CGFDN	91.9	95.0	81.5	88.7	85.0	90.6	88.4	79.8

[1] We choose ResNet18 as the backbone network for fair comparison.

TABLE V

AVERAGE COMPUTATION TIME (IN SECONDS) PER IMAGE ON THE TESTING SETS OF POTSDAM AND VAIHINGEN

Dataset \ Model	FCN-8s	UNet	SegNet	PSPNet	DeepLabv3+	PSANet	EncNet	S-RA-FCN	DANet	CGFDN
Potsdam	116.4	184.1	435.6	84.4	491.0	80.0	91.8	92.7	214.9	192.7
Vaihingen	15.4	24.5	58.2	11.4	65.6	10.9	12.3	12.4	29.2	25.8

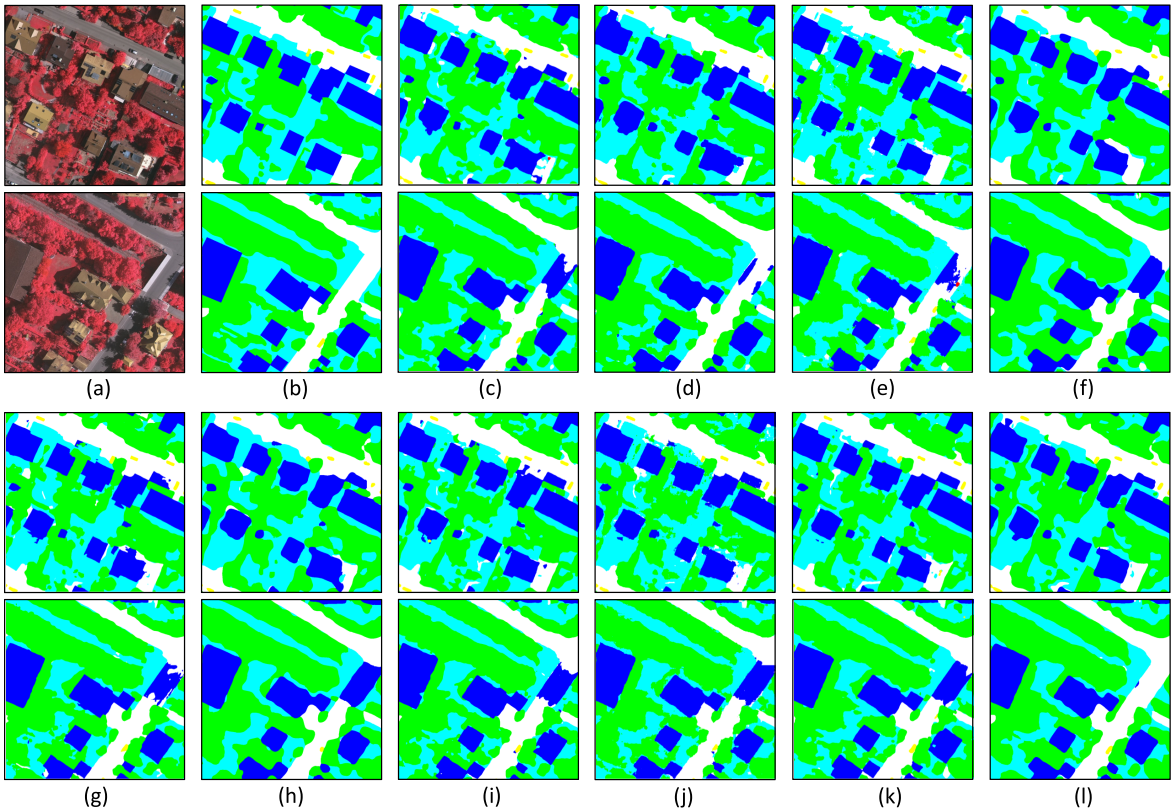


Fig. 7. Qualitative comparisons between different models using spatially contextual information on the Vaihingen data set. Six classes are included in the segmentation maps: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red). (a) Image. (b) Ground truth. (c) FCN-8s. (d) UNet. (e) SegNet. (f) PSPNet. (g) DeepLabv3+. (h) PSANet. (i) EncNet. (j) S-RA-FCN. (k) DANet. (l) CGFDN.

Furthermore, we also report the computation time of different models on two data sets in Table V. All experiments are performed using a personal computer with CPU of Intel Core i7-4790 and GPU of GTX TITAN X under the PyTorch framework. As shown in Table V, PSPNet, PSANet, EncNet, and S-RA-FCN cost less time than FCN-based models (i.e., FCN-8s, UNet, and SegNet) due to the use of lightweight modules for capturing contextual information. In comparison with them, DeepLabv3+ and DANet cost much more time, but they can obtain better segmentation performance in terms of OA and mean F1 score, shown in Tables III and IV. With respect to our proposed CGFDN, it not only saves computation time but also improves segmentation results compared with DeepLabv3+ and DANet.

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TABLE VI
QUANTITATIVE COMPARISONS WITH BENCHMARK METHODS ON THE POTSDAM DATA SET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1
SVL_1 [35]	83.5	91.7	72.2	63.2	62.2	77.8	74.6
DST_5 [42]	92.5	96.4	86.7	88.0	94.7	90.3	91.7
UZ_1 [9]	89.3	95.4	81.8	80.5	86.5	85.8	86.7
RIT_L7 [43]	91.2	94.6	85.1	85.1	92.8	88.4	89.8
UFMG_4 [1]	90.8	95.6	84.4	84.3	92.4	87.9	89.5
CVEO [44]	91.2	94.5	86.4	87.4	95.4	89.0	91.0
KLab_2 [45]	89.7	92.7	83.7	84.0	92.1	86.7	88.4
CAS_Y3 [46]	92.2	95.7	87.2	87.6	95.6	90.1	91.7
BUCTY5 [47]	93.1	97.3	86.8	87.1	94.1	90.6	91.7
CGFDN	92.6	97.0	86.8	87.6	94.9	90.8	91.8

TABLE VII
QUANTITATIVE COMPARISONS WITH BENCHMARK METHODS ON THE VAIHINGEN DATA SET

Method	Impervious Surface	Building	Low Vegetation	Tree	Car	OA	Mean F1
SVL_3 [35]	86.6	91.0	77.0	85.0	55.6	84.8	79.0
DST_2 [42]	90.5	93.7	83.4	89.2	72.6	89.1	85.9
UZ_1 [9]	89.2	92.5	81.6	86.9	57.3	87.3	81.5
RIT_L8 [43]	89.6	92.2	81.6	88.6	76.0	87.8	85.6
ONE_7 [26]	91.0	94.5	84.4	89.9	77.8	89.8	87.5
ADL_3 [48]	89.5	93.2	82.3	88.2	63.3	88.0	83.3
DLR_10 [21]	92.3	95.2	84.1	90.0	79.3	90.3	88.2
UFMG_4 [1]	91.1	94.5	82.9	88.8	81.3	89.4	87.7
CVEO [44]	90.5	92.4	81.7	88.5	79.4	88.3	86.5
CGFDN	91.8	94.7	83.1	89.7	82.9	91.0	88.4

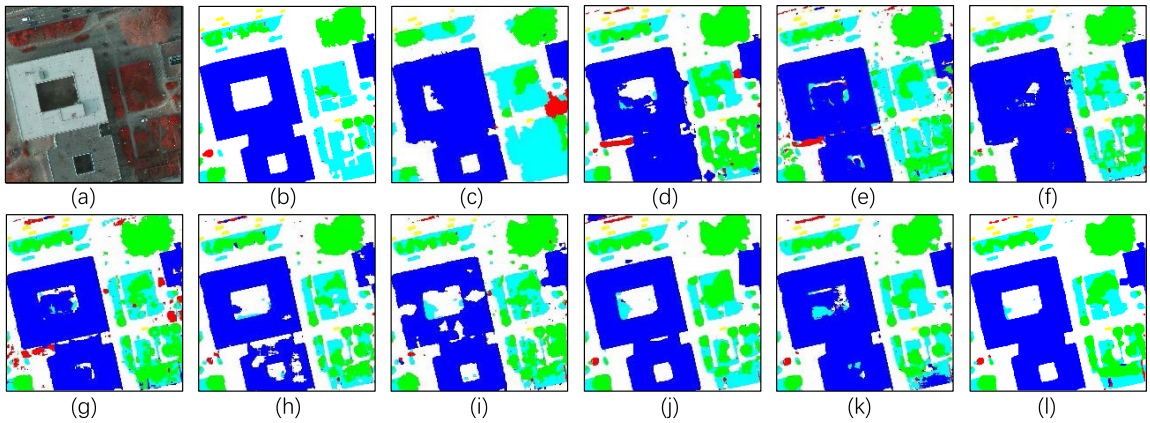


Fig. 8. Qualitative comparisons with benchmark methods on the Potsdam data set. Six classes are included in the segmentation maps: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red). (a) Image. (b) Ground truth. (c) SVL_1. (d) DST_5. (e) UZ_1. (f) RIT_L7. (g) UFMG_4. (h) CVEO. (i) KLab_2. (j) CAS_Y3. (k) BUCTY5. (l) CGFDN.

E. Comparison With Benchmark Models

Many benchmark models performed on Potsdam and Vaihingen have been published on the challenge evaluation website[4]. To further verify the performance of the proposed CGFDN, we compare CGFDN with several state-of-the-art benchmark models, including SVL_* [35], UZ_1 [9], DST_* [42], RIT_L* [43], UFMG_4 [1], CVEO [44], KLab_2 [45], CAS_Y3 [46], BUCTY5 [47], ADL_3 [48], ONE_7 [26], and DLR_10 [21]. In this section, CGFDN takes a composite image as the input. For fair comparison, we use the experimental results provided by the challenge evaluation website.

[4]http://www2.isprs.org/commissions/comm3/wg4/results.html

1) Performance Comparison: As shown in Figs. 8 and 9, the comparison models cannot distinguish confusing man-made objects and precisely locate fine-structured objects simultaneously. Especially, they easily confuse the building and the road and obtain inaccurate localization of the car. The main reason is that they focus on exploring spatially contextual information via convolutional features, which is insufficient to provide useful clues to discriminate different objects, especially for isolated objects. After integrating co-occurrence relations between different classes of objects into the convolutional features, CGFDN is able to obtain sharper and clearer boundaries on buildings and perform coherent and accurate labeling for cars. Meanwhile, as can be seen in Tables VI and VII, the OA and mean F1 scores obtained by CGFDN are higher than those of other benchmark models.

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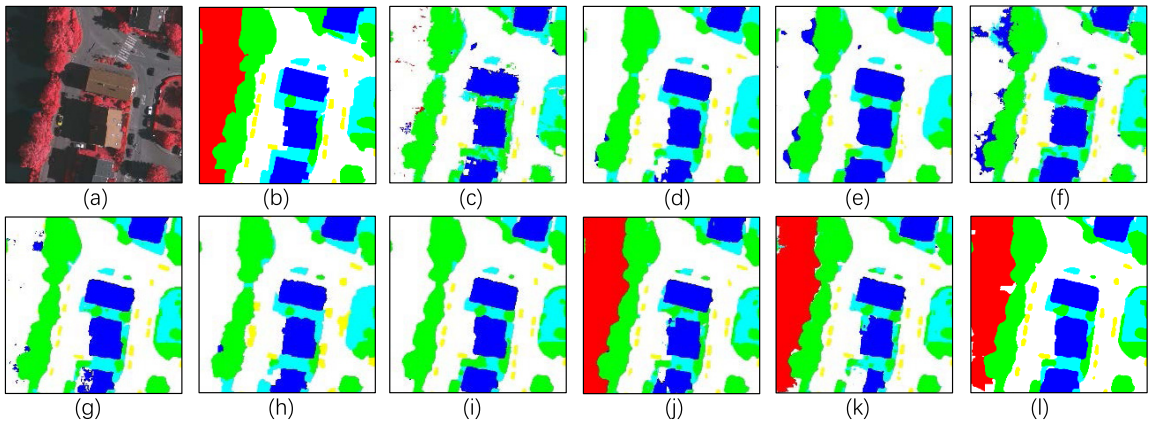


Fig. 9. Qualitative comparisons with benchmark methods on the Vaihingen data set. Six classes are included in the segmentation maps: impervious surface (white), building (blue), low vegetation (cyan), tree (green), car (yellow), and background (red). (a) Image. (b) Ground truth. (c) SVL_3. (d) DST_2. (e) UZ_1. (f) RIT_L8. (g) ONE_7. (h) ADL_3. (i) DLR_10. (j) UFMG_4. (k) CVEO. (l) CGFDN.

IV. Conclusion

In this article, we propose a novel CGFDN to fully exploit co-occurrence information for improving the performance of airborne image segmentation. Especially, we encode the co-occurrence relations between different classes of objects with the guidance of label information to enhance the discriminative ability of the convolutional feature. Compared with the original spatially contextual information, the enhanced one is more reasonable and effective. Due to the use of the co-occurrence information, CGFDN is able to make interclass features more distinguishable and intraclass features more consistent, obtaining more coherent and accurate segmentation maps. Experimental results show that the proposed CGFDN achieves better performance than several state-of-the-art methods on two challenging benchmarks: Potsdam and Vaihingen. In the future, we will design a lightweight feature decoupling module to replace the multibranch structure and deal with more challenging tasks, such as fine-grained segmentation of ultrahigh-resolution satellite images (e.g., Gaofen Image Data Set [49] consisting of 15 categories of land-cover objects).

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